

A Technical Introduction to Autonomous vehicle

Section 1

Privacy Some ACs require an internet connection to function, opening the possibility that a hacker might gain access to private information such as destinations, routes, camera recordings, media preferences, and/or behavioral patterns, although this is true of any internet-connected device.

Section 2

Espionage In 2018 and 2019, former Apple engineers were charged with stealing information related to Apple's self-driving car project. In 2021 the United States Department of Justice (DOJ) accused Chinese security officials of coordinating a hacking campaign to steal information from government entities, including research related to autonomous vehicles. China has prepared "the Provisions on Management of Automotive Data Security (Trial) to protect its own data". Cellular Vehicle-to-Everything technologies are based on 5G wireless networks. As of November 2022, the US Congress was considering the possibility that imported Chinese AC technology could facilitate espionage. Testing of Chinese automated cars in the US has raised concern over which US data are collected by Chinese vehicles to be stored in China and any link with the Chinese communist party.

Section 3

Development General Motors is developing the "Ultra Cruise" ADAS system, that will be a dramatic improvement over their current "Super Cruise" system. Ultra Cruise will cover "95 percent" of driving scenarios on 2 million miles of roads in the US, according to the company. The system hardware in and around the car includes multiple cameras, short- and long-range radar, and a LiDAR sensor, and will be powered by the Qualcomm Snapdragon Ride Platform. The luxury Cadillac Celestiq electric vehicle will be one of the first vehicles to feature Ultra Cruise.

Section 4

Software controls the vehicle, and can provide entertainment and other services. Over-the-air updates can deliver bug fixes and additional features over the internet. Software updates are one way to accomplish recalls that in the past required a visit to a service center. In March 2021, the UNECE regulation on software update and software update management systems was published.

Section 5

Driving systems Advanced driver-assistance systems (ADAS) automate specific driving features such as Forward Collision Warning (FCW), Automatic Emergency Braking

(AEB), Lane Departure Warning (LDW), Lane Keeping Assistance (LKA) or Blind Spot Warning (BSW). An ADAS requires a human driver to handle tasks that the ADAS does not support. ADAS contrasts to an automated driving system (ADS), which would be classified by SAE J3016 as Level 3 or higher.

Section 6

Ford In February 2024, a driver using the Ford BlueCruise hands-free driving feature struck and killed the driver of a stationary car with no lights on in the middle lane of a freeway in Texas. In March 2024, a drunk driver who was speeding, holding her cell phone, and using BlueCruise on a Pennsylvania freeway struck and killed two people who had been driving two cars. The first car had become disabled and was on the left shoulder with part of the car in the left driving lane. The second driver had parked his car behind the first car presumably to help the first driver. The NTSB is investigating both incidents.

Section 7

In July 2020, Toyota started public demonstration rides on Lexus LS (fifth generation) based TRI-P4 with Level 4 capability. In August 2021, Toyota operated a potentially Level 4 service using e-Palette around the Tokyo 2020 Olympic Village. In September 2020, Mercedes-Benz introduced world's first commercial Level 4 Automated Valet Parking (AVP) system named Intelligent Park Pilot for its new S-Class. In November 2022, Germany's Federal Motor Transport Authority (KBA) approved the system for use at Stuttgart Airport. In September 2021, Cruise, General Motors, and Honda started a joint testing programme, using Cruise AV. In 2023, the Origin was put on indefinite hold following Cruise's loss of its operating permit. In January 2023, Holon announced an autonomous shuttle during the 2023 Consumer Electronics Show (CES). The company claimed that the vehicle is the world's first Level 4 shuttle built to automotive standard.

Section 8

Safety model A safety model is software that attempts to formalize rules that ensure that ACs operate safely. IEEE is attempting to forge a standard for safety models as "IEEE P2846: A Formal Model for Safety Considerations in Automated Vehicle Decision Making". In 2022, a research group at National Institute of Informatics (NII, Japan) enhanced Mobileye's Reliable Safety System as "Goal-Aware RSS" to enable RSS rules to deal with complex scenarios via program logic.

Section 9

Level 4 – High automation In 2024, companies such as Waymo began offering robotaxi services in parts of the US with fully autonomous vehicles without safety drivers. These services all operate at a loss as of 2025, with

operating costs of about \$4.5–\$5.5/km (\$7–\$9 per mile), compared to \$0.6/km (\$1 per mile) for personal cars. The consultancy McKinsey estimated that bringing costs down to less than \$1.2/km (\$2 per mile) would take until 2035. In April 2023 in Japan, a Level 4 protocol became part of the amended Road Traffic Act. ZEN drive Pilot Level 4 made by AIST operates there.

Section 10

SAE Level 2 features are available as part of the ADAS systems in many vehicles. In the US, 50% of new cars provide driver assistance for both steering and speed. Ford started offering BlueCruise service on certain vehicles in 2022; the system is named ActiveGlide in Lincoln vehicles. The system provided features such as lane centering, street sign recognition, and hands-free highway driving on more than 130,000 miles of divided highways. The 2022 1.2 version added features including hands-free lane changing, in-lane repositioning, and predictive speed assist. In April 2023 BlueCruise was approved in the UK for use on certain motorways, starting with 2023 models of Ford's electric Mustang Mach-E SUV. Tesla's Autopilot and its Full Self-Driving (FSD) ADAS suites are available on all Tesla cars since 2016. FSD offers highway and street driving (without geofencing), navigation/turn management, steering, and dynamic cruise control, collision avoidance, lane-keeping/switching, emergency braking, obstacle avoidance, but still requires the driver to remain ready to control the vehicle at any moment. Its driver management system combines eye tracking with monitoring pressure on the steering wheel to ensure that drivers are both eyes on and hands on. Tesla's FSD rewrite V12 (released in March 2024) uses a single deep learning transformer model for all aspects of perception, monitoring, and control. It relies on its eight cameras for its vision-only perception system, without use of LiDAR, radar, or ultrasound. As of January 2024, Tesla has not initiated requests for Level 3 status for its systems and has not disclosed its reason for not doing so.